

12th July- This is the refined approach that i am talking now-
testing= Inch locking + specific SKUs? Or just the specific SKUs

Main aim- Machine allocation should be prefect, SKUs go to optimised machines, so that other SKUs can also built - and we get the perfectly optimised KPIs in the building and curing side.

1. Flrst we are starting with the current running presses, and current running stage1 and stage2 building machines and then we are mapping the SKUs with the running presses and machines. .
2. Now, if building machines have still some time left, then they will search for the secondary SKU as per the circular dependency time calculations.
3. If still some machines are left assigning then first we will check if the current SKUs that are in curing needs extra machines if their consumption is not fulfilled by the current assigned machines then first give priority to the those SKU's, if still some SKUs exists then we will assigning the remaining SKUs to them- by our discussed max. Ratio formula, and then we will search the secondary and tertiary SKU on that machine if the time still left- low consumption by the building machine- after checking and verifying the circular dependency calculations.
4. But, as mentioned in step3 approach- this is a completely dynamic process. Then, we are making a static curing consumption for 31 days at once - without considering the dynamic conditions, and if curing consumption will not include these SKUs and their qty, then building machines will not make them.
5. We are making a dynamic curing consumption schedule- 0 day building, then day1 building then day1 curing, then curing consumption for next day, then building schedule for next day and so on means curing consumption for next day is affected by the current day building and curing. Or just a static just based on day0? But, how will we incorporate this dynamic logic, day wise- discussed in point3? Or we are already implementing it?

6. Now, after assignment of RI SKUs, there me some press, some machines and SKU left- then we have to make curing CO and then we have to assign the remaining building machine to first if any RI SKUs needs it, current machines building that SKU not able to fulfill consumption otherwise assign as per the max. Ratio calculation that we discussed. Machine to specific SKU but make sure it's press is free, otherwise assign next best ratio SKU. Once the primary and secondary SKUs are decided machine will built these SKUs only until machine have some free time due to less consumption then search for a tertiary SKU or the demanded GT quantity of the current primary and secondary SKUs are not fulfilled.
7. For point 6 also we needs to calculate the curing consumption perfectly for each day dynamically. How we can do this? We are not currently doing this?
8. Similarly when the SKU demand from building and curing is completed. Then, we have to make CO on the building side as well as on the curing side as well. Now, we have to assign SKU based on this logic- Search for SKUs eligible for these curing press then we will decide machine wise the best SKU based on our max. Ration calculation formula, then secondary and tertiary if required- for swapping and then we will calculate time for the circular dependency and then we will start building.
9. For some starting days, I don't think machines need the secondary or tertiary SKUs because presses will not be free? If some presses will be free, then based on those press, select the SKUs with that max. Ratio formula but, if the current running other machines don't needs help- consumption is handled by those machines. Flrst priority to the current running machines help- so that no starvations occurs then search for other SKU- machine specific the time allows.
10. I think we should start building the stage1 output carcass 2 shift before- means in first s if the stage2 machines needs it in shift $s+2$. This is correct and feasible and stage2 will never go in starvation, because stage1 has 15 machines and stage2 has 6. Stage2

machines are fast so 2 shift buffer start is enough and sufficient as well.

11. Is curing a problem for us? We are not able to assigned the specific SKU to a specific machine, because press is not free? These conditions are also coming and does this creating any big and major issue for us?
12. And, we have to start assigning the machines to particular SKUs using this order- Stage2 then Unistage machine then BJ then VMI machines [600] first and then VMI (700)? I hope this is correct and optimised? ANY suggestion and comment here for deciding the order for the best possible KPIs?
13. Do you have proper clarity on this max ratio calculation formula and that is mathematically and technically perfectly ? Any suggestions and comments on that as well.
14. I needs one more suggestion should i go for the size locking on machine + assigning specific SKUs to machines? Or just the assigning specific SKUs to machines? Provide your recommendation after a prefect mathematical and technical analysis.
15. Main logic- if a machine has free time then, first must assign it to the RI SKU- if the consumption is higher and current machines are not able to fulfill it, then if free, calculate the max. Ratio from the discussed formula and then assign the SKU to that machine- [select only those SKUs those curing presses also free].